

Software Requirements Specification (SRS)

SauceDemo Web Application

Version: 1.0

Date: May 2026

Prepared by: QA Engineering Team

Project URL: <https://www.saucedemo.com>

Table of Contents

1. Introduction
 2. Overall Description
 3. User Classes & Test Accounts
 4. Functional Requirements
 5. Non-Functional Requirements
 6. Test Scenarios Overview
 7. Out of Scope
-

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the SauceDemo web application. It serves as the foundation for planning, designing, and executing test cases for the application.

1.2 Scope

SauceDemo is a demo e-commerce web application provided by Sauce Labs. It simulates a real-world online shopping experience including user authentication, product browsing, cart management, and a checkout process. The application is intentionally built with multiple user profiles that exhibit different behaviors, making it ideal for QA practice.

1.3 Definitions & Acronyms

Term	Definition
SRS	Software Requirements Specification
UI	User Interface
TC	Test Case
FR	Functional Requirement
NFR	Non-Functional Requirement
POM	Page Object Model
BDD	Behavior Driven Development

1.4 References

- Application URL: <https://www.saucedemo.com>
 - Selenium WebDriver Documentation
 - Cucumber BDD Documentation
-

2. Overall Description

2.1 Product Perspective

SauceDemo is a standalone web application that simulates an e-commerce store. It consists of the following pages:

Page	URL Path	Description
Login	/	Authentication entry point
Products	/inventory.html	Product catalog listing
Product Detail	/inventory-item.html	Single product details
Cart	/cart.html	Shopping cart contents
Checkout Step 1	/checkout-step-one.html	Buyer information form
Checkout Step 2	/checkout-step-two.html	Order summary & overview
Checkout Complete	/checkout-complete.html	Order confirmation

2.2 Operating Environment

- **Browser Support:** Chrome, Firefox, Edge, Safari
 - **Platform:** Web (Desktop & Mobile Responsive)
 - **Network:** Requires internet connection
 - **Automation Tool:** Selenium WebDriver + Cucumber (Java)
-

3. User Classes & Test Accounts

SauceDemo provides 6 built-in test users, each designed to simulate a specific behavior:

Username	Password	Expected Behavior
<code>standard_user</code>	<code>secret_sauce</code>	Full normal flow — all features work correctly
<code>locked_out_user</code>	<code>secret_sauce</code>	Login is blocked — error message displayed
<code>problem_user</code>	<code>secret_sauce</code>	UI bugs present — broken images, wrong form behavior
<code>performance_glitch_user</code>	<code>secret_sauce</code>	Simulates slow response / performance issues
<code>error_user</code>	<code>secret_sauce</code>	Some interactions throw errors
<code>visual_user</code>	<code>secret_sauce</code>	Visual/CSS layout bugs present

4. Functional Requirements

FR-01: Login / Authentication

Page: Login Page (/)

ID	Requirement	Priority
FR-01-01	The system shall display a username input field	High
FR-01-02	The system shall display a password input field	High
FR-01-03	The system shall display a "Login" button	High
FR-01-04	A valid username and password shall redirect the user to the Products page	High
FR-01-05	An invalid username or password shall display the error: <i>"Username and password do not match any user in this service"</i>	High
FR-01-06	An empty username field shall display the error: <i>"Username is required"</i>	High
FR-01-07	An empty password field shall display the error: <i>"Password is required"</i>	High
FR-01-08	The <code>locked_out_user</code> shall see the error: <i>"Sorry, this user has been locked out"</i>	High
FR-01-09	The error message shall include an (X) close button that dismisses it	Medium
FR-01-10	The system shall display the Sauce Labs logo and robot image on the login page	Low

FR-02: Product Catalog

Page: Products Page (</inventory.html>)

ID	Requirement	Priority
FR-02-01	The system shall display a list of 6 products	High
FR-02-02	Each product shall display: name, description, image, and price	High
FR-02-03	Each product shall have an "Add to cart" button	High
FR-02-04	Clicking "Add to cart" shall change the button text to "Remove"	High
FR-02-05	Clicking "Remove" shall revert the button back to "Add to cart"	High
FR-02-06	The shopping cart icon shall display a badge with the count of items added	High
FR-02-07	The system shall allow sorting products by: Name (A-Z), Name (Z-A), Price (Low-High), Price (High-Low)	Medium
FR-02-08	Clicking a product name shall navigate to the product's detail page	Medium
FR-02-09	The page header shall display the application title and a cart icon	High
FR-02-10	A hamburger menu (≡) shall be accessible in the top-left corner	Medium

FR-03: Product Detail

Page: Product Detail Page (`/inventory-item.html`)

ID	Requirement	Priority
FR-03-01	The system shall display the selected product's name, description, image, and price	High
FR-03-02	The system shall provide an "Add to cart" button	High
FR-03-03	Clicking "Add to cart" shall change the button to "Remove" and update the cart badge	High
FR-03-04	The system shall provide a "Back to products" link	High
FR-03-05	Clicking "Back to products" shall return the user to the Products page	High

FR-04: Shopping Cart

Page: Cart Page (`/cart.html`)

ID	Requirement	Priority
FR-04-01	The system shall display all items added to the cart	High
FR-04-02	Each cart item shall show: quantity, name, description, and price	High
FR-04-03	Each cart item shall have a "Remove" button	High
FR-04-04	Clicking "Remove" shall remove the item from the cart and update the cart badge	High
FR-04-05	The system shall display a "Continue Shopping" button	High
FR-04-06	Clicking "Continue Shopping" shall return the user to the Products page	High
FR-04-07	The system shall display a "Checkout" button	High
FR-04-08	Clicking "Checkout" shall navigate to Checkout Step 1	High
FR-04-09	An empty cart shall display no items (no error)	Medium

FR-05: Checkout — Step 1 (Information)

Page: Checkout Step 1 (</checkout-step-one.html>)

ID	Requirement	Priority
FR-05-01	The system shall display input fields for: First Name, Last Name, and Zip/Postal Code	High
FR-05-02	Clicking "Continue" with all valid fields shall proceed to Checkout Step 2	High
FR-05-03	Clicking "Continue" with an empty First Name shall show: <i>"First Name is required"</i>	High
FR-05-04	Clicking "Continue" with an empty Last Name shall show: <i>"Last Name is required"</i>	High
FR-05-05	Clicking "Continue" with an empty Zip Code shall show: <i>"Postal Code is required"</i>	High
FR-05-06	The system shall provide a "Cancel" button that returns the user to the Cart	High

FR-06: Checkout — Step 2 (Overview)

Page: Checkout Step 2 (</checkout-step-two.html>)

ID	Requirement	Priority
FR-06-01	The system shall display all items in the order with name, quantity, and price	High
FR-06-02	The system shall display the Item Total (subtotal before tax)	High
FR-06-03	The system shall display the Tax amount	High
FR-06-04	The system shall display the Total price (subtotal + tax)	High
FR-06-05	The system shall provide a "Finish" button	High
FR-06-06	Clicking "Finish" shall navigate to the Checkout Complete page	High
FR-06-07	The system shall provide a "Cancel" button that returns the user to the Products page	Medium

FR-07: Checkout — Complete (Confirmation)

Page: Checkout Complete (</checkout-complete.html>)

ID	Requirement	Priority
FR-07-01	The system shall display a success header: <i>"Thank you for your order!"</i>	High
FR-07-02	The system shall display a confirmation message and a Pony Express image	Medium
FR-07-03	The system shall display a "Back Home" button	High
FR-07-04	Clicking "Back Home" shall return the user to the Products page	High
FR-07-05	The cart badge shall be cleared (empty) after a successful order	High

FR-08: Navigation Menu

Accessed via: Hamburger icon (≡) on any page after login

ID	Requirement	Priority
FR-08-01	The menu shall contain: "All Items", "About", "Logout", "Reset App State"	High
FR-08-02	Clicking "All Items" shall navigate to the Products page	High
FR-08-03	Clicking "Logout" shall log the user out and redirect to the Login page	High
FR-08-04	Clicking "Reset App State" shall clear the cart and reset all button states	Medium
FR-08-05	Clicking "About" shall navigate to the Sauce Labs website	Low
FR-08-06	The menu shall close when the (×) button is clicked	Medium

5. Non-Functional Requirements

ID	Requirement	Category	Priority
NFR-01	The Login page shall load within 3 seconds on a standard connection	Performance	High
NFR-02	The Products page shall load within 3 seconds after login	Performance	High
NFR-03	All interactive elements (buttons, links) shall be clickable without scrolling on 1280×720 resolution	Usability	High
NFR-04	Error messages shall be clearly visible and descriptive	Usability	High
NFR-05	The application shall function correctly on Chrome, Firefox, and Edge	Compatibility	High
NFR-06	Password field shall mask the entered characters	Security	High
NFR-07	Users shall not be able to access <code>/inventory.html</code> without logging in first	Security	High
NFR-08	The cart count badge shall accurately reflect the number of items at all times	Data Integrity	High
NFR-09	Price totals on the checkout overview shall be mathematically correct	Data Integrity	High

6. Test Scenarios Overview

These are the high-level scenarios your test cases should cover:

Authentication

- Login with valid credentials → success
- Login with invalid credentials → error message
- Login with empty fields → field-specific error messages
- Login as locked-out user → locked error message
- Logout via hamburger menu

Product Catalog

- Verify all 6 products are displayed
- Sort products by all 4 sort options
- Add product to cart → badge updates
- Remove product from catalog → badge decrements

Shopping Cart

- Add multiple products and verify cart contents
- Remove item from cart
- Navigate back to shopping via "Continue Shopping"
- Proceed to checkout from cart

Checkout Flow

- Complete full checkout with valid data (happy path)
- Submit checkout form with empty fields

- Verify price calculations on order overview
- Verify cart is cleared after order completion

Navigation

- Open and close the hamburger menu
 - Reset App State clears the cart
 - Access all menu links
-

7. Out of Scope

The following are **not** in scope for this SRS as SauceDemo is a demo application:

- Payment processing (no real payment gateway exists)
- User registration / account creation
- Order history / past purchases
- Backend API validation
- Email confirmation flows
- Mobile native application testing
- Accessibility (WCAG) compliance testing